Data Analysis

Data analysis

- Digging deep into data that we have collected in the field / from an experiment.
  - Trends
  - Patters
- Explain a story from the data from
  - Behaviour
  - Practices
  - Culture
  - Lab experiments
  - System / user interaction

From the research

Approaches of data analysis

- Based research paradigms:
  - Ontology
  - Epistemology
- Positivists
  - Dominated by numeric data, to some extent, there can also subjective / text data
- Interpretivists
  - Dominated by subjective data / also have quantitative data

Quantitative data analysis

- Strategies
  - Descriptive data analysis
    - Basic statistical analysis
    - Techniques : tables, ratios, charts/graphs, averages and standard deviations
    - This is applicable when analysing single variable data. I,e, we do not seek to understand a relationship between two or more variables.
  - Correlation data analysis {inference data analysis}
    - Is used to analyse data with more than one variable where we see to establish a relationship between those variable.
    - Techniques include the T-Test and regression analysis

Qualitative data management

- The data may include: pictures, words, voices, videos etc.
- Approaches:
  - Deductive-based on pre defined concepts
  - Inductive- not constrained to predefined concepts
- Qualitative data is analysed using thematic data analysis
  - Establishing word, phrase, sentence that best describes pattern on a phenomenon under study
- Data analysis starts in the course of data collection
  - Reflections:
  - Take shorts of observation

Stages in qualitative data analysis

- Familiarisation
  - Check all data available
  - Notes are clear
  - Transcribing : converting your data from notes format to complete readable text. Essay– Transcripts
- Coding: a word, or phrase of our interest in relation to the study
  - E.g. Security : coded by color, line etc..
- Categorisation : Grouping related code
- Themes: develop these from the categories

Sampling

Sample

- A sample is a representation of a study population.
- Study population refers to the expected WHOLE participants in research.
- We are expected to involved everyone in a study:
  - Its expensive
  - Its time consuming
  - Hence, research used a sample of the participants
- Whatever size, but characteristically, the sample must be a true reflection of the population

Guidelines for selecting a sample

- Must be one that gives you adequate information about your study
- Must not be selected with biasness, hence, you should be clear how you selected them, e.g.
  - Purposive sampling
  - Random sampling
- Must accessible
- In quantitave : use formula to determine size
- In qualitative : information saturation and or information richness

Assignment

Write a structured research concept on any research problem of your choice in ICT field. This should be a 1-page work, excluding the cover page and references page.

The work should be structured as follows:

Cover page

Assignment page

- Background & Literature
- Problem
- Aim
- Objectives
- Study approach and research strategy
- Data Collection
- Data Analysis

Reference (Citation using APA 7th Edition)

Submission Guide:

- Date: 28th April 2026
- One soft copy in MS Word to *(https://drive.google.com/drive/folders/1JHPH0Ogz9Cqx0kxCWdj54DnUchDyB2P8?usp=sharing)*
- File name must PROG_GROUP_NAME_NAME_OF_SENDER E.G. *BISO_Group1_martinmsendema*
- One Hard Copy by 16:30Hrs
- Formatting: Arial 12, before 0, after 0, line spacing 1.5, jusify
- Separate for groupd for BIT, BIS, (BITW-BISW),(BIT-BIS-ODeL)
- 3 per group